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MT8676 Hypervisor Vcodec User Manual

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Version History

Version	Date	Author	Description
1.0	2024-10-28	Yougao.Liu	Official version
1.1	2024-11-11	Yougao.Liu	Added the scope of application for the video playback architecture in Section 1.3.1.2
1.2	2024-11-25	Yougao.Liu	Removed “Yocto+Android” in Section 1.3.1.2 and 1.3.1.3

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1 Basic Introduction to Video Encoding and Decoding

1.1 Purpose

The specific system architecture is described in this architecture design document to ensure that hardware resources can be effectively shared in a virtualized environment between Android and Yocto systems, providing users with a coherent and high-performance experience.

1.2 Scope

This document focuses on describing the high-level architecture of the system, covering design considerations from the hardware layer to the application layer, as well as communication and interaction among different components.

1.2.1 Abbreviation

Table 1-1. Abbreviation

Abbreviation	Explanation
AV1	AOMedia Video 1
CC	Cryptographic Checksum
CMD	Command
H.264	MPEG-4 Part 10, Advanced Video Coding, MPEG-4 AVC
HAL	Hardware Abstraction Layer
HDR	High Dynamic Range
Mbps	Megabits Per Second
OPP	Operating Performance Point
PS	Location Probability
SVP	Secure Video Playback
TS	Time Slot
V4L2	Video for Linux Two
VCodec	Video Codec
VCP	Video CoProcess
Vdec	Video Decoder
Venc	Video Encoder
VirtIO	Virt Input/Output

1.2.2 Performance

The video file formats currently supported in the system are shown in [Table 1-2](#).

Table 1-2. Video file format

Container	File Extension	8676 Android U0	Comment
MKV	".mkv", ".mka"	√	-
MP4	".mp4", ".m4a", ".m4v", ".3g2", ".3gp", ".3gpp", ".3gpp2"	√	-
PS	".mpg"	√	VOB is not supported
TS	".ts", ".m2ts", ".mts"	√	-
WEBM	".webm"	√	-
AVI	".avi"	X	Android S phase out
WMV	".wmv", ".asf"	X	Android S phase out
FLV	".flv", ".f4v"	X	Android S phase out
RMVB	".rmvb", ".rm"	X	-

Specific supported parameter list information for the video codec module is provided in Table 1-3.

Table 1-3. Vcodec performance

Category	Codec	Spec				
		Resolution	Frame Rate	Bit Rate	Bit Depth	Profile/Level
Decoder	H.264	4096x2176	60	160 Mbps	8/10	CBP, MP, HP, high 10 / 5.2
	H.265	4096x2176	60	160 Mbps	8/10	Main / 5.1 Main 10 / 5.1
	VP9	4096x2176	60	120 Mbps	8/10	Profile 0/2
	AV1	4096x2176	60	120 Mbps	8/10	Main/level5.1
Encoder	H.264	3840x2160	60	160 Mbps	8/10	BP, MP, HP; High 10 / 5.2
	H.265	3840x2160	60	160 Mbps	8/10	Main, Main10 / 5.1
Concurrency (HW capability)	-	4K60 VDEC + 4K60 VENC or equivalent throughput				

1.3 Architecture/Process Overview

1.3.1 Software Architecture

The architecture diagram related to video playback in the current system is the main focus of this chapter.

1.3.1.1 Android

The architecture of video playback in a single system is depicted in Figure 1-1.

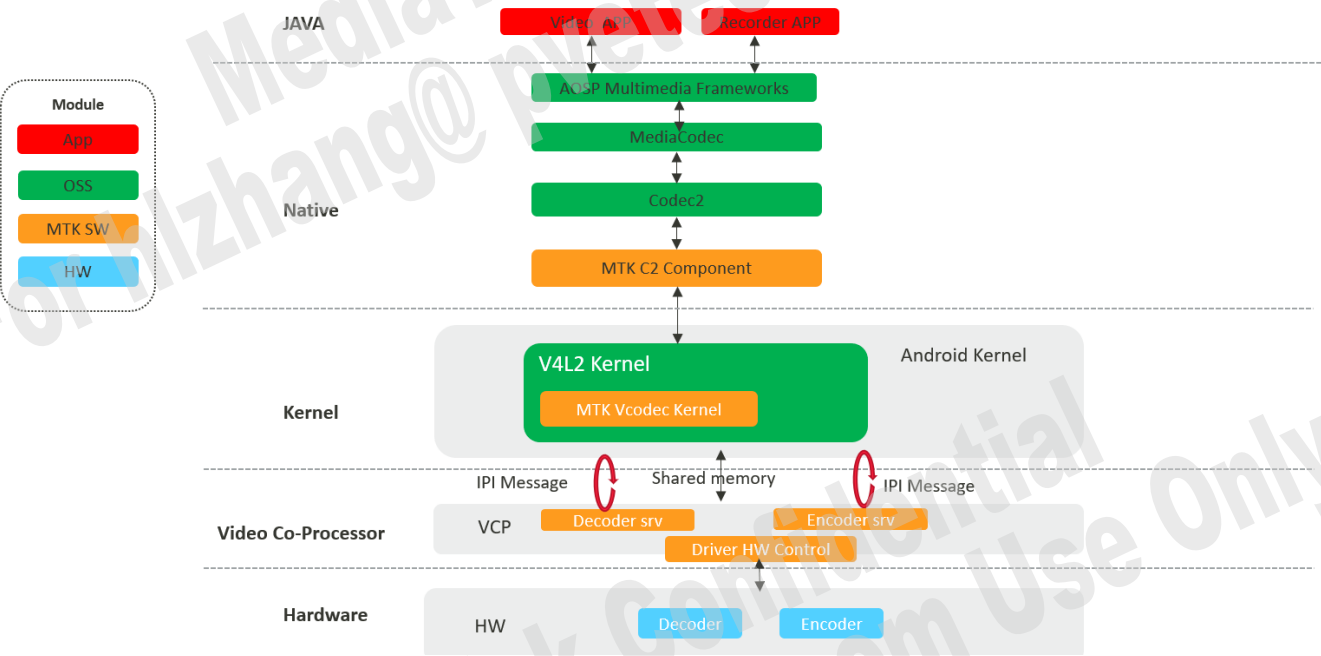


Figure 1-1. Android OS playback architecture

1.3.1.2 Hypervisor

The architecture of video playback in the Hypervisor system is shown in Figure 1-2. It is based on L(w/Vcodec)+A or L(w/o Vcodec)+L(w/o Vcodec)+A.

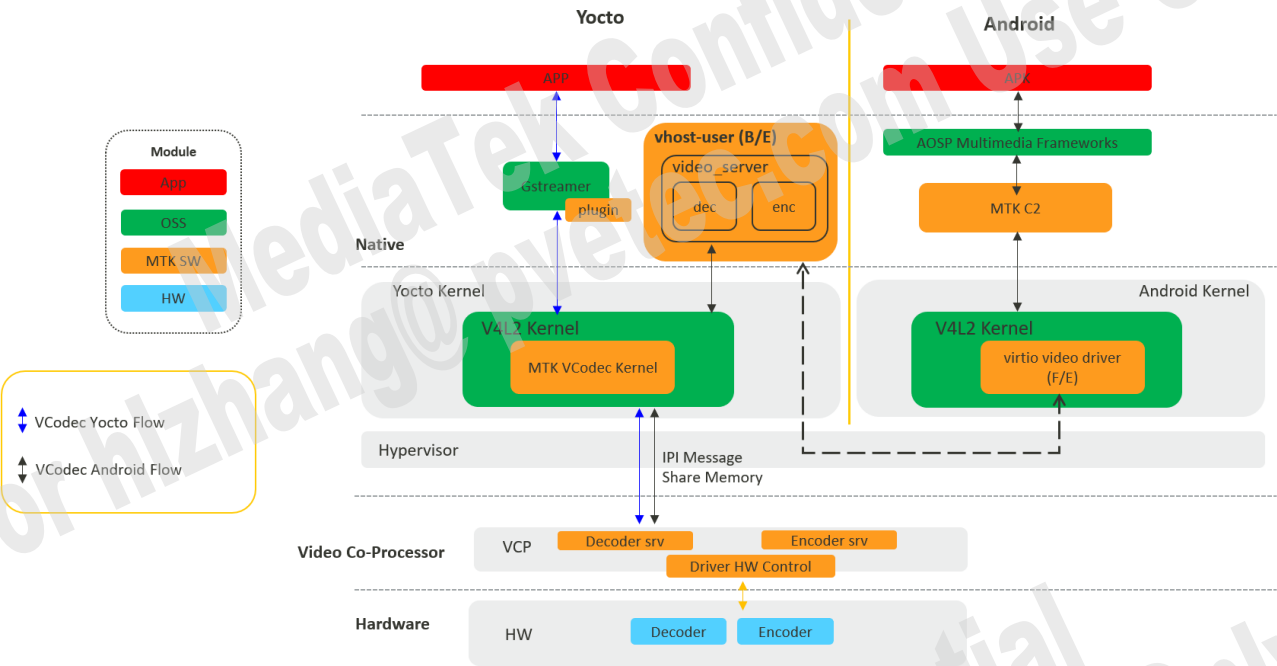


Figure 1-2. Hypervisor OS playback architecture

1.3.1.3 Hypervisor SVP

The architecture for SVP in the Hypervisor system is illustrated in Figure 1-3.

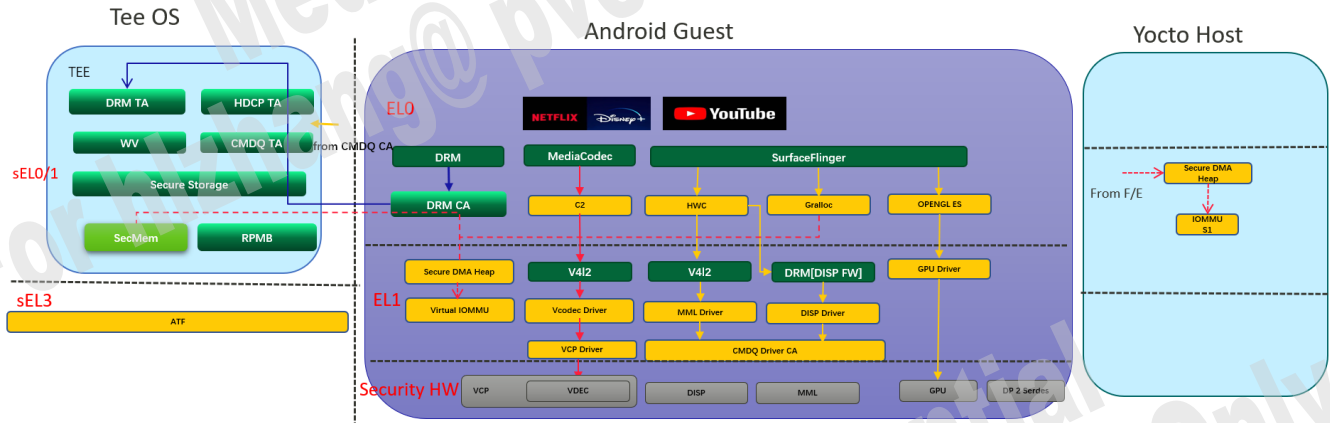


Figure 1-3. Hypervisor OS SVP playback architecture

1.4 Configuration/Customization Guide

1.4.1 Configuration of the Decoder Codec

The codec information supported by the decoder configured in the system is depicted in the following diagram.

```

112 |         </MediaCodec>
113 |         <MediaCodec name="c2.mtk.avc.decoder" type="video/avc">
114 |             <Alias name="OMX.MTK.VIDEO.DECODER.AVC" />
115 |             <Limit name="size" min="64x64" max="4096x2176" />
116 |             <Quirk name="wants-NAL-frgments" />
117 |             <Feature name="adaptive-playback"/>
118 |             <Limit name="concurrent-instances" max="15" />
119 |             <Limit name="performance-point-3840x2160" value="60" />
120 |         </MediaCodec>

```

1.4.2 Configuration of the Channel Supported by the Decoder

The information on the number of the channels supported by the decoder configured in the system is depicted in the figure below.

```

112 |         </MediaCodec>
113 |         <MediaCodec name="c2.mtk.avc.decoder" type="video/avc">
114 |             <Alias name="OMX.MTK.VIDEO.DECODER.AVC" />
115 |             <Limit name="size" min="64x64" max="4096x2176" />
116 |             <Quirk name="wants-NAL-frgments" />
117 |             <Feature name="adaptive-playback"/>
118 |             <Limit name="concurrent-instances" max="15" />
119 |             <Limit name="performance-point-3840x2160" value="60" />
120 |         </MediaCodec>

```

1.4.3 Configuration of Secure Decoder

The configuration information for secure decoders in the system is depicted in the following diagram.

```

<MediaCodec name="c2.mtk.avc.decoder.secure" type="video/avc">
  <Alias name="OMX.MTK.VIDEO.DECODER.AVC.secure" />
  <Limit name="size" min="64x64" max="1920x1088" />
  <Quirk name="wants-NAL-frgments" />
  <Feature name="secure-playback" required="true" />
  <Feature name="adaptive-playback"/>
  <Limit name="concurrent-instances" max="15" />
  <Limit name="performance-point-1920x1088" value="30" />
</MediaCodec>

```

1.4.4 Configuration of the Encoder Codec

The configuration information corresponding to the supported encoders in the system is depicted in the following diagram.

```

<MediaCodec name="c2.mtk.avc.encoder" type="video/avc" >
  <Alias name="OMX.MTK.VIDEO.ENCODER.AVC" />
  <Limit name="size" min="160x128" max="3840x2176" />
  <Limit name="alignment" value="16x16" />
  <Limit name="block-size" value="16x16" />
  <Limit name="concurrent-instances" max="10" />
  <Limit name="bitrate" range="1-160000000" />
  <Limit name="performance-point-3840x2160" value="60" />
  <Limit name="performance-point-1920x1079" value="180" />
  <Limit name="performance-point-1920x1080" value="120" />
  <Feature name="encoding-statistics" />
</MediaCodec>

```

1.4.5 Configuration of the Channel Supported by the Encoder

The configuration information of the number of channels supported by the encoder in the system is depicted in the following diagram.

```

<MediaCodec name="c2.mtk.avc.encoder" type="video/avc" >
  <Alias name="OMX.MTK.VIDEO.ENCODER.AVC" />
  <Limit name="size" min="160x128" max="3840x2176" />
  <Limit name="alignment" value="16x16" />
  <Limit name="block-size" value="16x16" />
  <Limit name="concurrent-instances" max="10" />
  <Limit name="bitrate" range="1-160000000" />
  <Limit name="performance-point-3840x2160" value="60" />
  <Limit name="performance-point-1920x1079" value="180" />
  <Limit name="performance-point-1920x1080" value="120" />
  <Feature name="encoding-statistics" />
</MediaCodec>

```

1.4.6 Configuration of Secure Encoder

The configuration information for the secure encoder supported in the system is depicted in the following diagram.

```

<MediaCodec name="c2.mtk.hevc.encoder.secure" type="video/hevc" >
  <Limit name="size" min="160x128" max="3840x2160" />
  <Limit name="alignment" value="16x16" />
  <Limit name="block-size" value="16x16" />
  <Limit name="quality" range="0-100" default="50" />
  <Limit name="concurrent-instances" max="10" />
  <Feature name="bitrate-modes" value="VBR,CBR,CQ" />
  <Limit name="performance-point-3840x2160" value="30" />
  <Feature name="special-codec" required="true" />
</MediaCodec>

```

1.4.7 Customization of the Playback Feature

1.4.7.1 HDR

Specific project customization support is required for HDR functionality.

1.4.7.2 SVP Playback

Customized support for the SVP feature requires specific project customization.

1.5 Common Issues/Troubleshooting

1.5.1 MTK C2_hal CMD

C2MtkComponent:

All C2MTK ALOGV logs of C2MtkComponent should be enabled

```
adb shell "setprop vendor.mtk.c2.enable.comp.log 2"
```

C2MtkVdec:

Enable C2MTK ALOGV logs of C2MtkVdec

```
adb shell "setprop vendor.mtk.c2.enable.vdec.log 2"
```

C2MtkVenc:

Enable C2MTK ALOGV logs of C2MtkVenc

```
adb shell "setprop vendor.mtk.c2.enable.venc.log 2"
```

C2MtkBufferManager:

Enable C2MTK ALOGV logs of C2MtkBufferManager

Enable C2MTK ALOGV internal logs of C2MtkBufferManager

```
adb shell "setprop vendor.mtk.c2.enable.bm.log 2"
```

```
adb shell "setprop vendor.mtk.c2.enable.bm.internal.log 1"
```

VCodec:

All debug level logs for VCodec should be enabled.

Enable debug level logs by <TAG> filter. E.g. V4L2Device

```
adb shell "setprop vendor.mtk.c2.enable.vcodec.log.V4L2Device 4"
```

```
adb shell "setprop vendor.mtk.c2.enable.vcodec.log 4"
```

```
adb shell "setprop vendor.mtk.c2.enable.vcodec.log.<TAG> 4"
```

1.5.2 MMDVFS CMD

Clock is forced to a specific step by OPP.

param1: 0→Vcore 1→vmm

OPP 0: highest clock rate

```
adb shell "echo 0 0 > /sys/module/mtk_mmdvfs_debug/parameters/force_step"
```

Vote clock to the specific step by OPP:

```
adb shell "echo 0 0 > /sys/module/mtk_mmdvfs_debug/parameters/vote_step"
```

Dump current clk:

```
adb shell "echo fmeter > /proc/clkdbg ; cat /proc/clkdbg | grep -e vdec -e cam"
```

Dump MMDVFS OPP:

```
adb shell "cat /proc/mmdvfs/mmdvfs_opp"
```

1.5.3 VCodec CMD

Vcodec log Enable/Disable

1: Enable

2: Disable

```
adb shell "echo 1 > /sys/module/mtk_vcodec_dec_v2/parameters/mtk_vcodec_dbg"
```

Control the log level for Vcodec.

codec_log: Common codec log level (0~31)

vpud_log: Vcodec wrap log level (0~7)

job_log: Job log level (0~7)s

```
adb shell "echo -codec_log 7 -vpud_log 3 -job_log 3 > /sys/module/mtk_vcodec_dec_v2/parameters/mtk_vdec_vcp_log"
```

1.5.4 Video Dump Feature

Input dump for the decoder (bit-stream for decoding)

The dumped file is saved in \data\vendor\vcodec\vdec_input_<timestamp>_<instance>.bs

E.g., \data\vendor\vcodec\vdec_input_20200716T133851.521_0xE80D6000.bs

```
adb shell "setprop vendor.mtk.c2.vdec.dump.input 1"
```

Dump of decoded video frames from the decoder output.

The dumped file is saved

in \data\vendor\vcodec\vdec_output_<timestamp>_<instance>_W<width>H<height>.yuv

E.g., \data\vendor\vcodec\vdec_output_20200716T133851.521_0xE80D6000_W1280H720.yuv

```
adb shell "setprop vendor.mtk.c2.vdec.dump.output 1"
```

Dump of decoded video frames from the decoder after color conversion.

The dumped file is saved in:

\data\vendor\vcodec\vdec_output_cc_<timestamp>_<instance>_W<width>H<height>.yuv

E.g., \data\vendor\vcodec\vdec_output_cc_20200716T133851.521_0xE80D6000_W1280H720.yuv

```
adb shell "setprop vendor.mtk.c2.vdec.dump.output.cc 1"
```

A binary debugging log containing decoder profiling data has been dumped.

The dumped file is saved in \data\vendor\vcodec\vdec_profiling_<timestamp>_<instance>.dat

E.g., \data\vendor\vcodec\vdec_profiling_20200716T133851.521_0xE80D6000.dat

```
adb shell "setprop vendor.mtk.c2.vdec.profiling 1"
```

1.5.5 Black Screen

- Verify if the video file is functioning correctly.
- Ensure that the video decoding module is started up normally.

```
adb shell "setprop vendor.mtk.c2.enable.vdec.log 2"
```

Confirm in the logcat logs if there is information similar to the following:

```
CCodec: allocate(c2.mtk.vp9.decoder)
C2MtkComponentStore: find: name=c2.mtk.vp9.decoder, key=c2.mtk.vp9.decoder
C2MtkVdec: [0xB400007D63859720] IntfImpl: name=c2.mtk.vp9.decoder
```

- Confirm that the decoding data is normal.
- Confirm that the decoding module has received the normal video stream data:

```
adb shell "setprop vendor.mtk.c2.vdec.dump.input 1"
```

- Is the GPU forced to be OK?

```
adb shell service call SurfaceFlinger 1008 i32 1
```

- The display module's functionality is confirmed to be normal.

Refer to the display document to confirm the display logic.

If all the above points are satisfactory, the detailed log analysis for issue localization needs to be performed by the Video playback owner.

1.5.6 Screen Distortion

- Confirm that the decoding data is normal.
- Confirm that the decoding module has received the normal video stream data:

```
adb shell "setprop vendor.mtk.c2.vdec.dump.input 1"
```

- Is the GPU forced to be OK?

```
adb shell service call SurfaceFlinger 1008 i32 1
```

- The display module's functionality is confirmed to be normal.

Refer to the display document to confirm the display logic.

If all the above points are satisfactory, the detailed log analysis to identify the issue needs to be performed by the video playback owner.

1.5.7 Occurrence of Lagging or Freezing

- Verify if the video file is functioning correctly.
- Ensure the video decoding module is started up normally.

```
adb shell "setprop vendor.mtk.c2.enable.vdec.log 2"
```

Confirm in the logcat logs if there is information similar to the following:

```
CCodec: allocate(c2.mtk.vp9.decoder)  
C2MtkComponentStore: find: name=c2.mtk.vp9.decoder, key=c2.mtk.vp9.decoder  
C2MtkVdec: [0xB400007D63859720] IntfImpl: name=c2.mtk.vp9.decoder
```

- The display module's functionality is confirmed to be normal.

Refer to the display document to confirm the display logic.

If all the above points are satisfactory, video playback owner needs to carefully examine the log to locate the issue.

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